

Recent Artificial Intelligence Advances in Detection and Diagnosis of Sickle Cell Disease: A review

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Abstract— Sickle cell anemia is a genetic disease characterized by a genuine alteration of hemoglobin that leads to the emergence of sickle-shaped red blood cells. The clinical diagnosis of sickle cell disease is based on blood analysis tests such as electrophoresis, chromatography and isoelectric focusing, blood count scanning or peripheral blood smears. These analyses aim to identify the presence of the hemoglobin S (HbS) in the sickle cell patient's blood. Nonetheless, these approaches face limitations linked to processing delays, substantial expenses, the need for specialized expertise, and inaccessibility in developing countries, where the prevalence of sickle cell disease is substantial. Thus, artificial intelligence techniques, including medical image segmentation and machine learning algorithms, offer innovative solutions to address these challenges. This paper gives an overview of recent advances related to image segmentation, feature extraction as well as classification approaches used in the detection and diagnosis of sickle cell disease. We introduced both computer vision techniques, machine learning and convolutional neural networks for sickle cell detection in microscopic images of blood smear. We discuss specific challenges related to segmentation methods while processing microscopic images containing overlapping cells. In addition, we examined challenges related to the robustness of convolutional neural network models used for feature extraction and then classification of sickle cells images. To conclude this paper, research prospects that can be explored in the future related to sickle cell detection in medical concerns are given.

Keywords—sickle cell detection, segmentation, classification, convolutional neural network, diagnostic.

I. INTRODUCTION

Sickle cell disease (SCD) is a genetic disease that affects the hemoglobin present in red blood cells [1]. When a person has SCD, the hemoglobin beta-globin gene is altered and the red blood cells change shape and become sickle-shaped when deoxygenated [2]. Polymerization and sickling of red blood cells leads to permanent creation of cells called sickled cells [3]. This morphologic alteration triggers cell destruction and shortens their lifespan, leading to hemolytic anemia [4]. Symptoms

associated with sickle cell disease include anemia, episodes of pain, hand and foot syndrome, increasing infections, failure to thrive, and vision problems [5]. These symptoms also call for several complications, including stroke, acute chest syndrome, pulmonary hypertension, leg ulcer [6].

SCD is one of the most common and widespread hereditary hematological disorders in the world [7] with approximately 300,000 babies that are born with sickle cell disease each year [8]. SCD exhibits a higher prevalence in Africa, ranging from 10% to 40% [9]. In sub-Saharan Africa alone, it is estimated that approximately 75% to 80% of people are carriers of the sickle cell gene [9]. Regarding Senegal, about 10% of the population is affected by SCD and each year, approximately 1,700 infants are delivered with the homozygous SS form [10]. Since 2006, the World Health Organization (WHO) has recognized sickle cell disease as a global public health problem [11].

The main challenge in diagnosing sickle cell disease lies in the accurate detection and counting of sickle cells due to their varied shapes and tendency to overlap in microscopic images [12]. The traditional approach used by healthcare professionals to detect sickle cell disease involves manually inspecting blood smear samples from patients under a microscope. As described in [13]–[15], this process is time-consuming and can be prone to errors due to overlapping red blood cells during the observation process. In addition, it requires the presence of a qualified hematologist, which may be difficult to find in some areas affected by the disease.

The use of artificial intelligence (AI) techniques, such as image processing methods and convolutional neural network models, offers fresh opportunities for enhancing sickle cell disease detection. These approaches are able to efficiently analyze microscopic images, including overlapping cells, allowing for more accurate detection and counting of sickle cells [16]. In recent years, AI has generated great enthusiasm in

medical research and has found numerous applications [17]. Due to enhanced computing power, vast data availability, and progress in machine learning, AI has become a valuable assistive technology in many fields, particularly in medical imaging [18]. Work has been conducted in the field of healthcare to highlight the positive impacts of using AI on the diagnosis, prevention, and treatment of various diseases [19]. The AI techniques are increasingly integrated into clinical diagnosis and treatment of hematological diseases, including sickle cell disease [20].

This review aims to explore AI techniques that emphasize image segmentation methods and machine learning algorithms, to precisely detect and diagnose SCD using microscopic images of affected patients. In addition, we highlight the performance limitations of these computer-based approaches when it comes to SCD detection.

The rest of the paper is organized as follows. Section 2 provides a state-of-the-art examination of recent artificial intelligence methods for sickle cell disease diagnosis. Current challenges identified in sickle cell disease detection are presented in Section 3. Finally, section 4 summarizes the paper and gives avenues of research to explore in perspective.

II. ARTIFICIAL INTELLIGENCE METHODS FOR DETECTING SICKLE CELL DISEASE

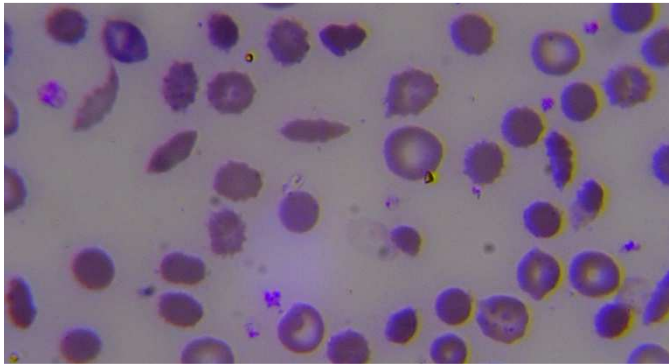


Fig. 1. Microscopic blood sample image of a patient with sickle cell disease

The traditional approach of identifying and counting sickle cells used in healthcare concerning diagnosing SCD is time-consuming, expensive and requires the intervention of a specialized physician. A tedious task that calls for several propositions based on AI techniques that aim to improve the manual processes of diagnosing SCD. The sickle cell disease detection methods based on AI techniques can be grouped into three categories: image segmentation and feature extraction methods, classification methods based on machine learning algorithms, and deep learning (DL) based SCD detection. This section reviews works related to SCD in the scientific literature. It provides an overview of recent studies and advances in the medical field, it highlights the strengths and weaknesses of each SCD detection method using microscopic images. All proposals abstracted in this paper are widely based on the analysis of datasets containing microscopic images of patients with SCD, as illustrated in Fig. 1. In Fig. 1., we can observe the shape and

arrangement of blood cells used to train AI algorithms to detect and diagnose SCD.

The state-of-the-art reviews were conducted using Google Scholar to identify primary literature in sickle cell disease detection. The proposals published from 2015 were included in these papers, without any language restrictions. In addition, databases were also integrated into the exploration strategy, including PubMed, Scopus, ArXiv, Embase. The following terms were used as keywords: artificial intelligence, machine learning, deep learning, convolutional neural networks, image segmentation, feature extraction, sickle cell disease, red blood cell, hematology, diagnosis, detection, classification.

The selected papers were chosen based on criteria related to:

- the publication date of the paper (the latest from 2015)
- the used artificial intelligence approaches (image segmentation, features extraction)
- the metrics used to evaluate the paper in terms of performance (accuracy, sensitivity, execution time)
- the usefulness of the proposed method when it comes to diagnosis and treatment of sickle cell disease

A. Segmentation and Feature Extraction Methods

Detecting and counting sickle cells in blood smear microscopic images is challenging for sickle cell disease diagnosis due to their varied shapes and potential overlap in the images. The methods used by healthcare professionals to identify and count cells have several limitations. To automate this process with improvement in accuracy, various image segmentation and feature extraction techniques have been applied to diagnose SCD. This part focuses on image segmentation and features extraction methods that have been used on blood smear microscopic images.

Menika et al. [21] proposed a method to detect and count sickle cells using image processing techniques and fractal dimension. The first step in this method was to collect blood smear images using a camera connected to a microscope. Then, a preprocessing step was undertaken, consisting of applying a median filter to the original images in order to reduce noise. After that, the red blood cells are segmented through edge detection algorithm and K-means algorithm to identify normal cells and sickled cells. The shape features of sickle cells are extracted based on their fractal dimensions to distinguish them from other cell shapes. In a final step, bounding boxes are used to detect and count sickle cells present in the image. This study shows that the image segmentation techniques used as well as the fractal dimension provide good results for the detection and counting of sickle cells. However, the edge detection algorithm is less effective in dealing with overlapping cells.

Chintawar et al. [22] proposed an image processing technique involving denoising and segmentation to discern the shape and size of red blood cells in sickle cell patients' blood. This method makes it possible to identify and quantify sickle cells among the 12 types of cells examined in the blood of a sickle cell patient. The proposed workflow consists of first acquiring microscopic images in RGB format from sickle cell

patients. This collection step is followed by the preprocessing step, where the original images are converted to grayscale. The grayscale images are filtered to attenuate impulse noise. Then, the edge detection algorithm is used to segment the cells in these images. This segmentation technique helps to define the boundaries of each cell so that cell borders, structures and contours are distinguished in the image. Finally, features such as circularity and color combined with the number of cells for each type are calculated. This framework can identify and assess the percentage of each cell type by considering the distinctive features of each type among the 12 cell types under scrutiny. However, the drawback was that the method is unable to deal with overlapping cells efficiently and the number of features used to categorize a cell is limited.

Abduraheem et al. [23] use two segmentation algorithms, such as Circular Hough Transform (CHT) and Watershed algorithm to automatically detect sickle cell disease. Both algorithms were implemented to extract circularity features from cells and determine the shape of overlapped cells in the images. A dataset of 233 microscopic images from sickle cell patients was used in this study. The CHT algorithm was more efficient when it comes to computing delay to distinguish normal cells from sickle cells in images. It requires a processing time of 8.71 seconds to classify cell types, while the Watershed algorithm takes 11.74 seconds to perform the same task. These methods were suggested for automating sickle cell disease diagnosis, and the findings demonstrate their significant potential for hematologists.

Sharma et al. [24] proposed a segmentation and feature extraction method with the aim of distinguishing normal cells, sickle cells, dracocytes and ovalocytes using the watershed segmentation algorithm. They collected 100 sample images of blood cells in which four classes were identified. The quality of the acquired blood smear microscopic images is first improved by the median filter algorithm. The watershed algorithm is then used to segment cells and extract morphological features, such as the metric value, aspect ratio and radial signature of each cell. Thus, extracted features are subjected to the KNN classifier to distinguish the four cell types, which had an accuracy of 80.6%. This method has the advantage of detecting cell types other than sickle cells. However, the Watershed algorithm requires more computation time because it requires filling in all local minima in the image to effectively separate overlapping cells.

Bheem et al. [25] proposed a method that involves image preprocessing, segmentation, feature extraction and classification of red blood cells in order to diagnose SCD. In the preprocessing phase, RGB images from the erythrocyte IDB1 [26] dataset are converted to grayscale, and the median filter is utilized to eliminate noise and enhance image quality. In the segmentation step, the authors first used Otsu thresholding to separate the cell type from the rest of the image. Then, we employed the Watershed segmentation algorithm to separate overlapping cells and assign labels to each cell type. Morphological closing eliminates undesired objects in segmented images. Following this, cell features, encompassing geometric and statistical attributes, are extracted, including circularity, aspect ratio, eccentricity, Standard Deviation, Variance, Entropy, and energy. Finally, the extracted features are fed into classifiers for the classification of the three cell

types. The RF classifier exhibited 92% accuracy in classifying the three cell types using morphological features, surpassing SVM, RF, and RL algorithms. This study demonstrates that combining Otsu thresholding and the Watershed algorithm efficiently handles overlapping cells, enabling the extraction of additional cell properties. However, the Watershed algorithm requires a significant amount of computing time to effectively separate overlapped cells, which makes this method expensive.

In the work of Mohamed et al. [26], the Sobel filter is used to extract the shape of globules and determine whether the diameter of a cell is less than 0.8 μ m. The aim of this research is to identify and differentiate between sickle cells and normal cells. This approach has proven highly valuable, especially in remote areas with limited medical access. However, it does not specifically address the segmentation of overlapping cells, potentially leading to incorrect detections.

B. Classification Methods Based on Machine Learning Algorithms

The morphological features extracted from the blood cells were used to classify sickle cells and other cell types in the microscopic images using algorithms such as KNN (k-nearest neighbors), SVM (support vector machine), NB (naive Bayes), RL (logistic regression), DT (decision tree) deep learning algorithms. Most of the extracted features from red blood cells consist of the cell ratio aspect, circularity, entropy, variance, eccentricity. These quantitative features then serve as an input to classification algorithms to detect sickle cells and distinguish them from other forms of red blood cells. This approach which fuses image segmentation techniques and classification algorithms, was used to effectively detect and classify SCD with good performance. Sharma et al. [24] used the KNN classifier to classify sickle cells, dracocytes, ovalocytes, and healthy erythrocytes using the aspect ratio metric values and radial signature of each cell as input variables. It achieves a classification accuracy of 80.6% and a sensitivity of 86.4%. The results show the KNN used is able to preserve cell information in the learning phase. However, the KNN classifier requires enormous computing time for large datasets and it is very sensitive to outliers.

Sagar Yeruva et al. [27] compare SVM, KNN, RL and RF classifiers accuracy while proposing a framework that classifies SCD. The authors develop classifiers to dissociate sickle cells from normal and thalassemia cells. They collected 1387 files of sickle cell patients. Based on collected data, 13 features are identified to train the classifiers. The SVM, KNN, RL and RF classifiers had accuracies of 76%, 77%, 78%, 95% and 98% respectively. The results show that the RF algorithm achieves the best prediction accuracy in detecting sickle cell disease. However, the performance of the SVM and KNN algorithm is not optimal when applied to the test data.

Bheem Sen et al. [25] used four machine learning algorithms to classify normal cells, sickle cells, and other cell types of cells using features extracted from each type. Variables used to train the classifiers include circularity, aspect ratio, eccentricity, deviation, variance, entropy, and energy. The objective of this research is to create a system that can effectively recognize and categorize red blood cells, even when they are superimposed. After training the algorithms, results

showed that RF obtained the best performance with an accuracy of 92%. The Naïf Bayes (NB) algorithm obtains the lowest precision with 88%. We also see that the RF adapts better to the training data, recording sensitivity rates of 95%, 92% and 91% in class 1, 2 and 3, respectively.

Alagus et al. [28] proposed a method for early identification of sickle cell disease based on the Inception V3 model and the SVM and KNN classifier from blood smear microscopic images. The aim of this method is to facilitate the diagnosis of sickle cell disease, particularly in children and women. The images used in this study are collected from the public “erythrocytes IDB”[29] database. The Inception V3 image recognition model is used to extract features of normal cells and sickle cells from images. The MO-BGWO optimization algorithm is used to select relevant features. Then, KNN and SVM classifiers are used to classify normal cells and sickle cells based on the extracted features. Results give an accuracy of 99% and a sensitivity of 98%. The results of this study could contribute to the early detection of sickle cell disease and prevent possible serious complications.

Multilayer perceptron (MLP) was included in the work of Yeruva et al. [30] for the diagnosis of sickle cell disease. This model was subjected to validation which showed 99% accuracy when classifying normal, sickle and thalassemia cells on the validation dataset. The performance of the MLP classifier was then compared to that of the SVM, KNN, RL, and RF classifiers to evaluate the relative performance of each algorithm.

C. Extraction and Classification Methods Based on Deep Learning

Recent work has mainly focused on the use of deep learning for efficient detection of sickle cells. DL techniques are used for both feature extraction and classification of red blood cells from blood smear microscopic images. The DL methods such as convolutional neural networks and recurrent neural networks have been shown to be very effective in extracting and accurately classifying sickle cell disease medical images.

Mengjia Xu et al. [31] proposed a method based on convolutional neural networks to classify red blood cells into eight classes, namely discocytes, echinocytes, elongated, granular, ovalocytes, reticulocytes, drepanocytes and stomatocytes. In this study, the authors first extracted patches of globules from different categories in the microscopic images. The sizes of the extracted patches are normalized or a mask image for each class is generated at the output. Then, a convolutional neural network is created to classify the blood cells. In the last step of this work, the authors analyzed the shape factor of each type of red blood cell in order to evaluate the shape parameters of the studied classes. This method gives an accuracy of 87.50% and an area under the ROC curve (AUC) of 0.94. However, it is important to note that to achieve even better results, this method would require more precise segmentation and training on a larger dataset.

Mo Zhang et al. [32] implemented a method for simultaneous segmentation and classification of red blood cells based on the U-Net architecture with the introduction of deformable layers. This method aims to segment the blood cells and classify them into three types: normal cells, sickle cells and

other types of cells. This method is able to efficiently segment cells present in shaded areas of microscopic images. The results showed that the proposed model is able to classify cells with 99% accuracy. The comparative evaluation also showed that this method outperformed the standard U-Net approach and region growing method in terms of semantic segmentation of sickle cells. However, a noteworthy constraint in this work is the small size of the training dataset used to train the model.

Alzubaidi et al. [33] proposed three deep learning models to detect sickle cells from blood smear microscopic images. In this study, the authors used a dataset consisting of 626 blood smear images divided into three classes: 202 images of normal cells, 211 of sickle cells and 213 images composed of other cell shapes. To address the issue of insufficient data, the authors employed data augmentation techniques (flip, luminosity) and transfer learning. The training model achieved a classification accuracy of 99.98%.

In the study by B. Sen et al. [34], five pre-trained models such as ResNet50, ResNet101, VGG16, VGG19, InceptionV3 were used to classify red blood cells into three categories: normal cells, sickle cells and other cells types. The images used in this method are collected from the erythrocyteIdb1[29] dataset. Flipping data augmentation techniques were used to enrich the training samples. From the results obtained, the InceptionV3 model obtains the best performance with a classification accuracy of 91%, surpassing the ResNet50, ResNet101, VGG16 and VGG19 models. We note that ResNet and VGG require more training data to better classify cells than InceptionV3, which explains their relatively low performance.

Another broader method of diagnosing hematological diseases in order to improve patient care was proposed by Karagül et al. [35]. The authors used convolutional neural networks to classify 14 types of anemia. Around 1477 files of patients diagnosed with anemic disorders were used in this experiment. The Convolutional neural network (CNN) model had an overall accuracy of 84.63% after 100 epochs.

Emmanuel G. Dada et al. [36] used deep learning techniques to classify a patient's red blood cells into normal, sickle, and neonatal cells. This study aims to lower the expenses associated with sickle cell disease diagnosis by presenting a more cost-efficient approach. In this endeavor, five models built upon CNN architecture were tested to identify sickle cell disease in microscopic blood smear images. Using the applied data augmentation techniques, the PCNN-15 and DAPN-48 models achieved 99% and 100% accuracy ahead of the PCNN-48, VGG 19 and ResNet-50 models, respectively. Depending on the geographical locale in which this investigation is undertaken, the outcomes could wield substantial influence on the timely detection of sickle cell disease and subsequently curtail the financial resources earmarked for its therapeutic intervention.

P. Balamanikandan and al. [37] introduced an approach that relies on mathematical modeling involving quantum graph theory and a Long Short-Term Memory (LSTM) neural network for discriminating between sickle cells and normal cells. This technique aims to decrease the processing time and decision-making errors associated with the identification of sickle cell disease in patients. The process starts with the use of a DNA (Deoxyribonucleic Acid) sequence as input. The quantum graph

theory model is used to extract the characteristics of the elastic properties of each sequence. A set of eight features is extracted, and a spanning tree is generated to calculate the relational values. Then, the LSTM neural network is used to classify normal cells and sickle cells. The proposed DL model achieved a classification accuracy of 99.12%, endorsing the model's effectiveness.

Recent research findings show that DL models provide better performance when it comes to blood cell detection and classification tasks. [38].

Karunasena et al. [39] proposed a method for detecting sickle cell cells based on the Region-based Convolutional Neural Network (R-CNN) architecture. The given R-CNN architecture comprises two stages, including a feature extraction step and a classification step. CNN is applied for the extraction of relevant features from sickle cells. Subsequently, the features extracted from each region are fed into the SVM classifier to ascertain the object's class to which that region pertains. In addition, coordinates are estimated to enclose the object within the detected region using a bounding box. In this approach, 150 sample blood smear images were used to train the CNN model. The results indicated that the R-CNN network achieved 95.2% accuracy in sickle cell detection.

Ihtisham et al. [40] developed a model that uses convolutional neural networks, i.e. Faster R-CNN, to detect and categorize blood cells through microscopic images. This approach aims to automate the process of diagnosing hematological diseases that affect blood cells. The proposed model includes four steps: producing blood smears, collecting microscopic blood smears, labelling images and training the faster R-CNN model. The model is trained on annotated images to predict three classes of cells by delimiting each one with a bounding box. This method was compared with the manual SCD diagnostic approach to evaluate its performance. According to the results, the proposed model reaches up to 91.83%.

Maxime Darrin et al. [41] proposed a hybrid framework that incorporates convolutional and recurrent neural networks (RNN) layers to analyze video sequences of unbalanced red blood cells in sickle cell disease patients. The video sequences used were captured using a microscope and each video included between 6 and 100 images. This hybrids model provides an accuracy of 97%. The authors suggest that this research could be advantageous for monitoring the health status of SCD patients.

Goswani et al. [42] proposed an AI method with the goal of enhancing the manual identification and classification of sickle cells. The proposed method relies on using the ResNet50 deep learning model to accurately detect and categorize sickle cells within microscopic images derived from blood samples of patients with sickle cell disease. The trained ResNet50 model achieved an impressive accuracy of 93.88% on the tested dataset.

A comparative study was conducted by Kawuma Simon et al. [38] to assess the accuracy and performance of the pre-trained models employed in sickle cell disease detection. Experimental tests are based on five pre-trained models including VGG16, VGG19, Inception V3, ResNet50 and ReNet. All these models were evaluated on the same dataset to classify normal cells and

sickle cells from a microscopic image of a blood smear. The results show that the Inception V3 model is more suitable for the classification of sickle cells, and it obtains an accuracy of 100% ahead of the VGG16, VGG19, ResNet and ReNet models with an accuracy of 99%, 97%, 82 and 67%, respectively. We can see that the inceptionV3, VGG16, and VGG19 models yield good results, even though the performance has improved due to the increase in the number of filters. On the other hand, ResNet50 and ReNet models require more training data to learn more from extracted features. The performance of the ResNet50 and ReNet models could be improved by using a larger sickle cell dataset.

Niksa Praljak et al. [43] use deep learning to automate the sickle cell counting process to facilitate the analysis of microfluidic biomarker assay data. The given model was trained on 107 images of microfluidic biomarkers from patients with sickle cell disease. The model achieved an average uncertainty of 3%, assessed through accuracy metrics on the test dataset.

Another recent study, using the U-Net model, was proposed by Yasheng Chen et al. [44] for the detection of modular lesions in children and young adults with sickle cell disease. The U-Net model was trained on brain magnetic resonance images (MRI) from these two categories of patients. The U-Net model predicts the presence of a lesion with a sensitivity of 100% and an accuracy of 74%. This study is conducted to facilitate the clinical diagnosis of modular lesions.

Bhatia et al. [45] propose a new method based on a personalized neural network to classify discocytes cells, ovalocyte cells and sickle cells from microscopic images from patients with sickle cell disease. Of the 428 microscopic images collected, a dataset of 10,377 single-cell images was generated, on which the model was trained. In this study, the model achieved an accuracy of 81%. After comparison with the LightGBM classifier, Inception V3, ResNet, VGG6-16, VGG-19, ResNet-50, Fine-Tuned, DenseNet-121 and EfficientNetB0 models, the proposed model obtained the best accuracy.

Detection and classification of red blood cells from microscopic images are crucial steps in the automatic diagnosis of sickle cell disease. TABLE 1 highlights the performance, advantages and limitations of some methods detailed above and used by different researchers for the diagnosis of sickle cell disease. In table 1, we summarize the studies carried out over the last five years, which emphasis recent proposals related to the detection and diagnosis of SCD. This table illustrates recent advancements in enhancing sickle cell detection via image segmentation, machine learning, and deep learning techniques. It also underscores the constraints associated with each approach.

In addition to the research summarized in this paper, TABLE 1 provides a non-comprehensive compilation of notable studies underscoring the significance of advancements in sickle cell disease diagnosis facilitated by artificial intelligence techniques. The methods used by each author, the expectations in terms of performance and the limits of each of these studies are well described in TABLE 1. These various factors expedite our ability to draw perspectives for our future research.

These limitations should motivate researchers to enhance current methods and suggest systems for integration into the

sickle cell disease diagnosis process. Some of these limitations targeted here are widely discussed in section III and include the morphological variability of sickle cells, the quality of microscopic images, the treatment of overlapped cells, the lack of clinical dataset, and the clinical validation of data labeling process.

Besides reviewed proposals limitations, we propose paths to explore future research topics that address challenges around formulating effective strategies to address cell overlap in images, the limited access to comprehensive annotated datasets for researchers, and the integration of AI models into medical concerns when it comes to sickle cell disease diagnosis.

TABLE I. PERFORMANCE, STRENGTHS, WEAKNESSES OF REVIEWED SICKLE CELL DETECTION METHODS

Years	Methods	Performances	Strenghts	Weaknesses
2015	Segmentation using K-means [21]	Allow good detection and count of sickle cells	Significant time reduction in features extraction stage when training DL algorithms	Not taking into account overlapped cells in the sickle cells identification stage
2016	GrayScale, edge detection algorithm [22]	Enable to accurately count the percentage of cells on blood smear image	Good refining ability to highlight cell shape and edge from blood cell image	Unable to process overlapping cells.
2016	Watershed segmentation algorithm, KNN [24]	Detect blood cells with an accuracy of 80.6% and a sensitivity of 87.2%	Efficient to properly classify SCD	Fairly number of considered features when training the algorithm
2017	CHT and Watershed algorithm [23].	Optimum execution time	Significant time reduction in features extraction stage when training IA models	Liable to lead to a false diagnosis due to presence of leucocyte in blood smear
2017	SCD detection using CNN [31]	Allow SCD classification with an accuracy of 89.2% and an average AUC value of 0.97%	Make robustness in features extraction and classification in SCD diagnosis steps	Luck of accuracy when overlapped cells. Any information about the SCD stage
2017	Deformable U-Net [32]	High accuracy performance of 99% when classifying single-class red blood cells	Recommended to reduce noise on image, fixed heterogeneous shapes and overlapped cells	Does not access the number of cells which has a significant rule in SCD diagnosis
2020	CNN, transfer learning [33]	Accuracies up to 99%	Exhibit data augmentation and transfer learning strength to improve sickle cells classification	No pre-processing is done on the images before the feature extraction step and classification
2020	CNN, RestNet50 [43]	Robust detection ability even with poor images conditions	Better sensitivity in contrast to manual diagnostic method	Rely on a manually labeled training dataset, which can require considerable effort in terms of time and human resources
2021	SVM, KNN, RL, DT and RF [27]	RF outperform with accuracy result near 96% when it comes to SCD detection	Highly recommended for good precision when it comes to detect blood disorder including thalassemia	The classes on which the algorithms are trained are unbalanced, which can lead to an overfitting problem
2021	Classifier MLP [30]	High SCD prediction accuracy of 99%	Efficient to properly classify SCD	Give any information about the disease status
2021	Inceptionv3, ResNet and VGG [34]	Inception V3 91% outperform with prediction accuracy of 91%	Significantly reduce SCD diagnostic processing time	Give any information about the disease status
2021	RF, SVM, RL, NB [25]	RF reaches better performance with 92% of prediction accuracy when classifying cells	Strong ability to accurately identify cells type from blood smear image	Any comparison with similar image segmentation methods to validate its performance
2021	CNN-based SCD deection [35]	Good prediction accuracy of 84.63%	Good prediction while managing various types of anemia from blood smear image	Any information about a potential anemic disorder
2023	VGG19, RESNET-50, PCNN-15, PCNN-48 and DAPN-48 [36]	PCNN-15 and DAPN-48 give high average accuracy of 99.92% in SCD detection	Boost efficiency to discern between normal, neonatal and sickle cells	Any information about the type and the stage of the disease
2023	Quantum graph theory, LSTM [37]	Good accuracy rate up to 99.12%	Improve LSTM performances when dealing with features elasticity	Precision can be improved. High computing and memory cost
2023	CNN and RNN [41]	Accuracy of 97% when identifying sickle cells	Efficient identification ability to analyze noisy microscopic cell images from video sequences	—

2023	Inceptionv3, KNN and SVM [28]	Good performances score related to precision, accuracy, sensitivity, specificity and F1 score of 99%, 98%, 99%, 98% and 96% respectively	Good effectiveness to discriminate normal cells to sickle cells from blood smear	May need image processing step to improve performance
2023	SCD detection using R-CNN [39]	Good average accuracy of 95.20% with low error rate 0.5%	Good sensibility in sickle cells identification ability among others cells.	Can improve accuracy when help RCNN algorithm to process nested cells in the image
2023	Faster RCNN [40]	Suitable for hematological diseases diagnosis with accuracy near to 91.83%	Fit for hematological diseases diagnosis	Unable to effectively detect nested cells
2023	ResNet50 [42]	Good detection ability of 93.88% when it comes to detect sickle cells	Give highly contribution that help to process overlapped cells	Used dataset is restricted in size and diversity, which can potentially affect model sensitivity
2023	VGG16, VGG19, InceptionV3, ResNet and ReNet [38]	InceptionV3 gives efficient SCD detection with an accuracy of 97.3%	Give significant in using deep learning methods for the detection of sickle cells	—
2023	U-Net-based SCD detection [44]	U-Net showed 100% sensitivity in predicting the presence of SCD with an accuracy of 74%	Ability to identify with low error the spinal cord lesions in children and young with SCD	Although he recorded high sensitivity, its low accuracy can lead to a notable false positive decision
2023	Custom Deep CNN [45]	Custom Deep CNN outperform with an accuracy of 81% when classifying various erythrocyte morphologies	Strong ability to accurately classify various blood cells, even with overlapping sickle cell	Luck of diversity in dataset samples could affect the model's ability to handle broader morphological variations. Model need to be validated with a real-world clinical setting to evaluate its reliability in SCD diagnosis

III. DISCUSSION

The detection and accurate classification of sickle cells in red blood cells is a difficult job in diagnosis due to the variety and arrangement of cells in microscopic images. This study shows that the application of image processing techniques and Deep Learning methods could represent a promising way to facilitate the detection of sickle cell disease from microscopic images of blood smears. Despite the satisfactory results provided by recent work (section 2), it remains challenging to improve the accuracy of SCD detection and allow health experts to integrate intelligent systems into the daily diagnosis of this disease. This section outlines the challenges identified in this investigation, which may serve as potential areas for future research in the realm of sickle cell disease detection.

- The morphological variability of sickle cells: The diverse morphologies of sickle cells, which vary with the degree of anemia, can complicate the diagnosis. Collaborating with health professionals to accurately annotate cell images and categorize the various typical forms of targeted cells in SCD diagnosis can be undertaken. This work will make it possible to design more precise SCD identification models that would help doctors to better direct care based on anemia severity for each patient.
- The quality of microscopic images: Microscopic images may exhibit noise and artifacts due to equipment quality and image capture conditions. Indeed, the image aspect can make cell segmentation and therefore detection more complex and degrade the learning performance of models. A pre-processing step incorporating more

efficient denoising and image enhancement techniques can be used beforehand to further improve the quality of an image. Hence, improved images can aid in extracting more salient features that enhance the performance of models for sickle cell detection.

- Treatment of overlapped cells: Effective processing of overlapping cells still remains one of the major challenges in the accurate classification of sickle cells in microscopic images. Blood cells frequently overlap and therefore complicate segmentation and classification tasks on image. Investigating advanced techniques using neural network architectures like Vision Transformers [46] and deeper models may be advantageous for enhancing the segmentation of overlapped cells.
- Datasets: Standard datasets, such as erythrocytes IDB [29], are limited in size and diversity. For better performance, more comprehensive and diverse data collection initiatives, involving actual blood cell images obtained from healthcare facilities, could be pursued. Nevertheless, ethical and confidentiality concerns must be carefully considered.
- Clinical validation and generality of SCD detection models: One of the key challenges involves the development of standardized clinical protocols to assess the real-world efficacy of AI models in clinical practice. This entails gathering a collection of comprehensive and accurate medical datasets from patients with SCD as well as the creation of standard diagnostic criteria. Furthermore, it is crucial to conduct comparisons between the performance of AI models to that of

traditional clinical approaches, while enforcing the reliability and reproducibility of results. The clinical validation should encompass a diverse range of patient populations, all while respecting specific ethical and data confidentiality concerns.

IV. CONCLUSION AND FUTURE WORK

Sickle cell disease, also called sickle cell anemia, is one of the most common genetic diseases worldwide. Despite notable progress in symptom management, SCD remains still incurable. This paper offers an outline of recent developments in SCD detection through artificial intelligence techniques. We explore both image processing methods, machine learning and deep learning approaches aimed at enhancing the identification of sickle cells in microscopic blood smear images. Reviewed methods performances, strengths and limitations are discussed to highlight challengers and potential research paths to explore. The key challenges include frequently pertain to devising effective approaches for handling cell overlap in images, the restricted availability of extensive annotated datasets for researchers, and AI model integration in SCD diagnosis.

In addition, the issue of generalizing AI models in the diagnosis of SCD is paramount, given the global prevalence of this autosomal genetic condition. Subsequent research endeavors should aim to create an intelligent, real-time automated system for the abnormal cells detection and enumeration to support diagnostic and treatment decisions for sickle cell disease. Such a system could lead to a faster, more accurate, less expensive and more efficient process, while ensuring continuous monitoring of sickle cell patients.

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